

Siwon Lee

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EDUCATION

Washington University in St. Louis

Ph.D. in Economics

MO, USA

Aug 2024 – Present

- Research Fields: International Trade, International Finance, Asset Pricing, Portfolio Theory, Machine Learning

Washington University in St. Louis

M.A. in Economics

MO, USA

Dec 2025

Sungkyunkwan University

B.A. in Global Economics, B.S. Mathematics

Seoul, Korea

Mar 2021 – Aug 2024

Math: Real Anal., Diff. Geom., Prob. Theory, Basic Math. for AI, Comp. Applied Math.

Software: OS, Intro DB, Data Struct., Intro AI, Sys. Prog., Algorithms

Econ: Adv. Macro, Game Theory, Econometrics, Math. Econ., Intro Fin. Acct.

LUSEM & HANKEN University

Winter Joint Exchange Program

Sweden and Finland

Jan 2023

- Visited KPMG corporation in Helsinki and discussed sustainability
- Discussed critical thinking and examples of sustainability in each country

CONFERENCE PROCEEDINGS

A Pooled Exceedance Rate Hides Two Failures in Firm Earnings-at-Risk Under Trade Policy Uncertainty

Author

ACM ICAIF 2026 proceedings (forthcoming)

- Evaluated Earnings-at-Risk models on 402,780 firm-quarters under trade policy uncertainty
- Separated Gaussian tail misspecification from firm heterogeneity using quantile and conformal methods
- Improved coverage across firm sizes through size-specific calibration

Household Portfolio Choice under Trade Policy Uncertainty with a Deep Operator Network

Author

ACM ICAIF 2026 proceedings (forthcoming)

- Extended Merton portfolio choice with TPU-driven jump risk and stochastic equity volatility
- Learned the two-dimensional HJB solution operator using a multiple-input DeepONet
- Enforced short-sale and adding-up constraints exactly through a softmax portfolio head
- Reduced policy evaluation from 198 ms to 0.6 ms and compared allocations with AAII survey data

WORKING PAPERS

The Impact of Tariff and Trade Policy Uncertainty Shocks on Household Asset Portfolios

Author

Working paper

- Estimated stock, bond, and cash portfolio responses to TPU shocks using monthly local projections
- Modeled TPU shock transmission through firm inventories and imperfect inventory information
- Developed a variation of the Merton model for robust portfolio choice under Knightian uncertainty
- Characterized hedging demands against TPU shocks, ambiguity, and regime-transition risk

Domestic Firm Financial Vulnerability Consequences of Exchange Rate Depreciation

Author

Working paper

- Estimated firm financial vulnerability after exchange rate depreciation using local projections
- Identified heterogeneous effects of trade openness and financial hedging
- Solved nonconvex equilibrium conditions using Deep Equilibrium Networks
- Developed a structural model of endogenous exchange rates and heterogeneous firms

Homogeneity, Systemic Risk, and the Regulation of Automated Trading in India, the US, and the EU

Author

Working paper

- Modeled correlated automated trading, price feedback, and leveraged liquidation in continuous time
- Analyzed position limits, margin requirements, and circuit breakers with reopening auctions
- Compared regulation in India, the US, and the EU across market depth and automated trading growth

Lost in Translation or Strategic Choice? LLM Evidence on Multilingual Risk Disclosure Asymmetry

Co-author with Wonwoo Kim

Working paper

- Compared multilingual risk disclosures of US-listed Korean and Japanese firms using LLMs
- Documented persistent cross-language differences in financial, market, and governance risk disclosure
- Developed tests to detect and mitigate cross-language measurement bias in LLM text analysis

WORK IN PROGRESS

Market Impact of Large Shareholder Stock Sales: An Agent-Based Approach

Author

Draft

- Simulated liquidity and volatility responses to large shareholder stock sales
- Modeled heterogeneous investor behavior in limit order books using agent-based simulations
- Analyzed how information asymmetry and risk preferences amplify price impact and volatility
- Explored performative prediction feedback between investor forecasts, trading, and market prices

Labor Market Conditions and Household Financial Vulnerability

Author

Research Proposal

- Built a state-year panel of labor market indicators and mortgage delinquency
- Estimated links between labor market vulnerability and household financial distress
- Tested lagged specifications and robustness across model variations

CONFERENCE AND WORKSHOP PRESENTATIONS

2026 ZUEL Workshop on Frontiers in Finance Research

International workshop presentation

2026

OTHER PRESENTATION

The Effect of Internet Reviews on the Demand of Experience Goods: Case of Korean Movie Industry

Sungkyunkwan-KEIO Symposium

Jan 2024

Stabilizing ETF Price Gaps through Liquidity Provision with AI

2022 Capstone Design and AI Hackathon

Oct 2022

Forecasting Inflation and Economic Recession: Comparison of Econometrics and Machine Learning

2022 SKKU Annual Conference on Economics

Aug 2022

REVIEW EXPERIENCE

2026 ICAIF Main Conference

Reviewer

ACM

2026

21st Economics Graduate Student Conference

Reviewer

WashU, USA

Aug 2026

20th Economics Graduate Student Conference

Reviewer

WashU, USA

Aug 2025

COMPLETED RESEARCH PROJECTS

Analyzing the Impact of Monetary Policy on Construction and Facility Investment <i>Co-author</i>	SKKU, Korea Mar 2024 – Jul 2024
Analyzing the Effect of Monetary Policy on Technological Progress: NAICS 334 <i>Team Leader</i>	DB, Korea Dec 2023 – Feb 2024
Comparison of South Korea's 1998 and 2008 Economic Crises <i>Team Member</i>	SKKU, Korea Feb 2023 – Jun 2023
Numerical Investigation for Economic Modeling Using Python <i>Team Member</i>	SKKU, Korea Jun 2022 – Jul 2022
Analyzing the Relationship Between COVID-19 and Trade <i>Team Leader</i>	SKKU, Korea Apr 2022 – Jun 2022
Forecasting Changing Phase of Financial Market When Implementation CBDC <i>Co-author</i>	Korea Feb 2022
Implementation of CBDC to Reduce Public Antipathy Toward Privacy Violations <i>Co-author</i>	SKKU, Korea Aug 2021
Project of Convergence Basics <i>Team Member</i>	SKKU, Korea Jun 2021 – Aug 2021
Calculated Production Possibility Frontier <i>Team Member</i>	SKKU, Korea Apr 2021 – Jun 2021

TEACHING EXPERIENCE

Assistant Instructor (WashU, USA) • Introduction to Macroeconomics • Introduction to Microeconomics	Aug 2025 – Present
Teaching Assistant (SKKU, Korea) • Algorithm and Problem Solving • Computer Thinking and SW coding • Problem Solving	Feb 2023 – Jun 2024
Young Economic Education Supporters (Bank of Korea, Korea) <i>Lecturer</i>	Apr 2023 – Dec 2023
Sungkyun Tutoring (SKKU, Korea) • Intermediate Microeconomics Theory • Intermediate Macroeconomics Theory • Econometrics	Feb 2022 – Jun 2024

WORK EXPERIENCE

TRC Research Assistant <i>Research Assistant</i>	UHK, Remote Nov 2023 – Jan 2024
Wisdomagora <i>Editor Management Intern</i>	Seoul, Korea Sep 2022 – Feb 2023

EXTRACURRICULAR ACTIVITIES

Transdisciplinary Mentoring Community <i>Mentee</i>	WashU, USA Aug 2024 – Jul 2025
Global Leaders in Finance(GLIF) <i>Team Member</i>	Seoul, Korea Mar 2021 – Mar 2022
MIMIC <i>Team Member</i>	SKKU, Korea Sep 2022 – Jun 2024
Seekers <i>President</i>	SKKU, Korea Mar 2022 – Jun 2024

AWARDS

2022 SKKU Annual Conference on Economics	SKKU, Korea
<i>Top Prize</i>	<i>Aug 2022</i>
2022 Capstone Design and AI Hackathon	The Korean Association of Computer Education
<i>Encouragement Prize</i>	<i>Oct 2022</i>
Learning-Fair	SKKU, Korea
<i>College Dept. Dean's Awards</i>	<i>Nov 2021</i>
Sungkyun Tutoring	SKKU, Korea
<i>Excellence Prize</i>	<i>Dec 2022</i>
Facebook Hacker Cup 2021	Facebook, Online
<i>Round 2</i>	<i>Sep 2021</i>
Big Data Analysis Contest by NH Investment	NH Investment
<i>Participation Prize</i>	<i>Nov 2022</i>
Dean's List	SKKU
	<i>Aug 2022</i>

SCHOLARSHIP

Fellowship & Full Scholarship	Aug 2024 – Aug 2029
	<i>WashU</i>
Scholarship for Extracurricular Challenge	Jan 2024
	<i>SKKU</i>
Scholarship for Global Learning Leader	Dec 2023
	<i>SKKU</i>
Scholarship for Global Exchange Student	Mar 2023
	<i>SKKU</i>
Scholarship for Academic Excellence	Aug 2022
	<i>SKKU</i>
Scholarship for Academic Excellence	Feb 2022
	<i>SKKU</i>
State Scholarship	Feb 2021
	<i>Government of South Korea</i>

ADDITIONAL INFORMATION

Professional Memberships: American Finance Association

Achievements: ADsP, Certification of Cody Joint Event(MATLAB)

Skills: Programming(C, Python, MYSQL, MATLAB), LaTeX, MS Office(PowerPoint, Excel)

Languages: Korean(Native), English(Fluent)

Hobbies: Table Tennis, Tennis, Listening to Music, Basketball